

Detailed Local Development/Testnet Architecture

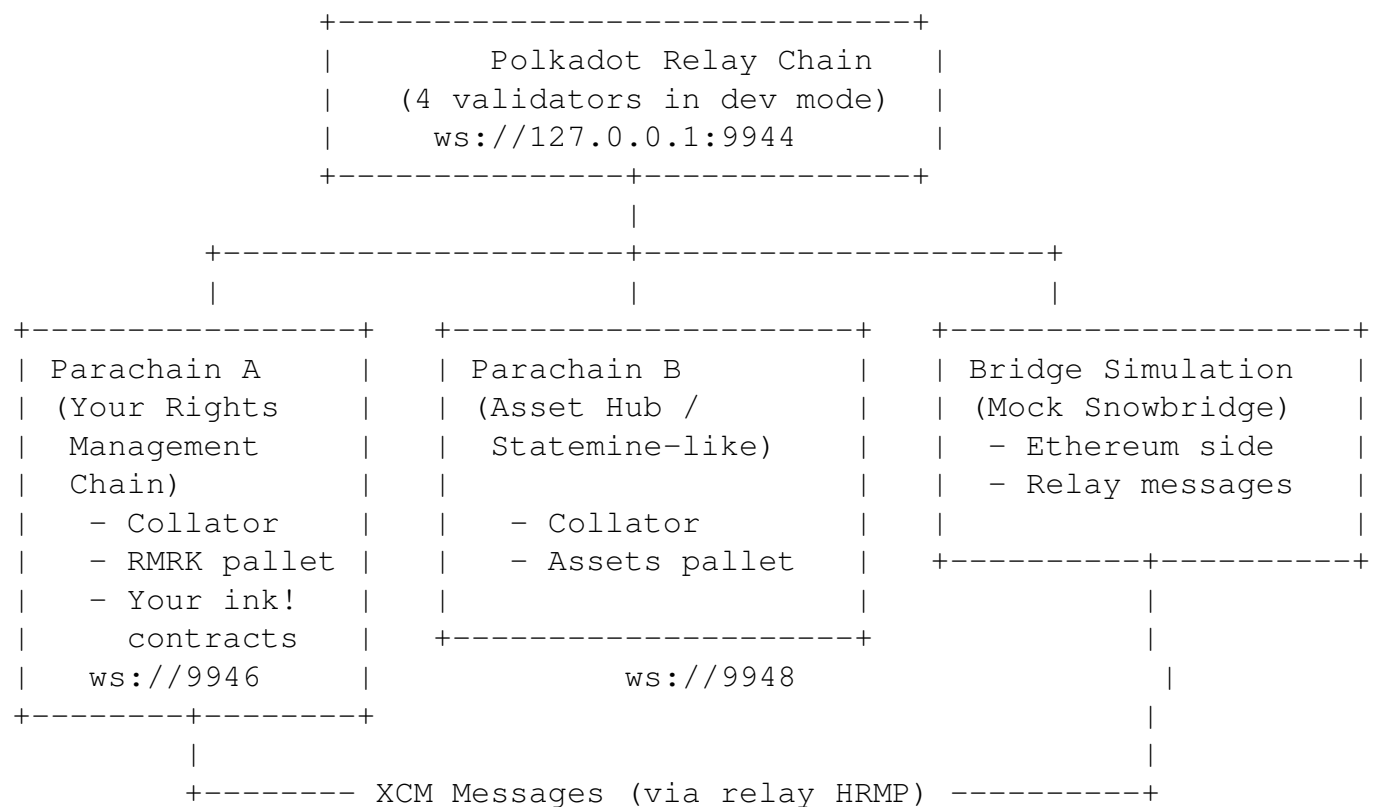
Current stack: ink! 6 + pallet-revive

Parachain A (Rights Management Chain) is the **content-rights-parachain** runtime with **pallet-revive** (ink! 6 / PolkaVM) and ink! 6 contracts. When using the single-parachain Zombienet config (`my-content-rights.toml`), the parachain RPC is **ws://127.0.0.1:9990**.

This is a detailed local development/testnet architecture for a cross-chain content rights management service. It exemplifies a practical configuration that can be executed on a single high-performance development machine. The objective is to replicate as comprehensively as possible the genuine Polkadot ecosystem locally, encompassing the relay chain, numerous parachains, RMRK collections, ink! contracts, and a fundamental bridge simulation.

Architecture Overview (Text Summary)

[Developer Machine / Local Cluster]



External Tools / Monitoring:

- Prometheus + Grafana (metrics)
- SubQuery / Subsquid (indexing events)
- Polkadot.js Apps (UI interaction)
- Local IPFS node (metadata storage)

Recommended Local Ports & Access Points

Table 1: Recommended Local Ports and Access Points

Component	Type	WS Port	HTTP Port	Purpose / Access URL
Relay Chain	Node	9944	9933	ws://127.0.0.1:9944
Parachain A (Rights Mgmt)	Collator	9946	9935	Main development target
Parachain B (Asset Hub)	Collator	9948	9937	Testing XCM asset transfers
Mock Bridge	Custom	9950	9951	Simulated Ethereum endpoint
Prometheus	Monitoring	—	9090	http://localhost:9090
Grafana	Dashboard	—	3000	http://localhost:3000
SubQuery Node	Indexer	—	3001	GraphQL endpoint for events
IPFS	Storage	—	5001	http://localhost:5001/api

Suggested docker-compose.yml Skeleton (high-level)

```
version: '3.8'

services:
  relay:
    image: parity/polkadot:v1.15.0
    command: --dev --tmp --rpc-external --ws-external --rpc-cors all
    ports:
      - "9944:9944"
      - "9933:9933"

  para-a-collator:
    image: parity/polkadot-parachain:v1.15.0
    command: --alice --collator --chain ./chainspecs/para-a.json
      --base-path /data --port 40333 --ws-port 9946
      --rpc-port 9935 -- --chain rococo-local
    depends_on:
      - relay
    ports:
      - "9946:9946"
      - "9935:9935"

  mock-bridge:
    build: ./bridge-simulator
    ports:
      - "9950:9950"
      - "9951:9951"

  prometheus:
    image: prom/prometheus:latest
    ports:
      - "9090:9090"
```

```
grafana:
  image: grafana/grafana:latest
  ports:
    - "3000:3000"
```

Implementation Notes (Phase 4–5)

Actual Architecture Deployed

The local development architecture evolved significantly beyond the original plan. Two distinct topologies were used:

Topology 1: XCM Testing (2-chain)

```
Rococo Relay Chain (alice, bob)
+-- ParaA (100) -- Content Rights at ws://127.0.0.1:9990
+-- ParaB (200) -- Consumer chain at ws://127.0.0.1:9991
```

Config: `zombienet-xcm-test.toml`, used for performance testing and XCM latency measurement.

Topology 2: Snowbridge Testing (4-chain + Ethereum)

```
Rococo Relay Chain (alice, bob)
+-- Bridge Hub (1013) at ws://127.0.0.1:8943
+-- AssetHub (1000) at ws://127.0.0.1:9910
+-- Content Rights (100) at ws://127.0.0.1:9990
  ^
Ethereum (Geth v1.17.1 at :8545 + Lodestar v1.35.0 at :9596)
+-- Gateway contracts (16 deployed)
+-- Beacon state service at :8080
+-- Beacon relay
+-- Ethereum relay
```

Config: `zombienet-snowbridge.toml`, used for Ethereum bridge E2E testing.

Changes From Original Plan

Table 2: Changes From Original Plan

Planned	Actual	Notes
4 relay validators	2 validators (alice, bob)	Sufficient for local dev
ParaA at ws://9946	ParaA at ws://9990	Port changed in config
ParaB as Asset Hub at ws://9948	ParaB as consumer chain at ws://9991	Separate AssetHub in 4-chain topology
Mock Snowbridge	Full Snowbridge v1	Geth + Lodestar + real relayers + 16 Gateway contracts
RMRK pallet	<code>pallet-nfts</code>	RMRK abandoned
SubQuery/Subsquid	Not used	Events queried directly via polkadot-js API
Prometheus + Grafana	Prometheus only	Scraped via <code>resource-monitor.mjs</code>
Local IPFS	Not used	Metadata hash stored on-chain
Docker for all components	Zombienet native provider	Simpler, faster startup

Setup Scripts Created

Table 3: Setup Scripts

Script	Purpose
<code>scripts/start-ethereum.sh</code>	Start Geth + Lodestar for local Ethereum
<code>scripts/deploy-gateway.sh</code>	Deploy 16 Snowbridge contracts via Forge
<code>scripts/snowbridge-full-setup.sh</code>	Full automated Ethereum bridge setup (~40 min)
<code>scripts/open-hrmp-channels.mjs</code>	Open HRMP between 2 parachains
<code>scripts/open-hrmp-snowbridge.mjs</code>	Open HRMP for 4-chain topology
<code>scripts/configure-snowbridge.mjs</code>	Configure gateway, Ether asset, accounts